

Computing Methods for Physics – 14 June 2023

Your exam material (code files, plots, etc.) must be submitted via google classroom by 18:00 as a single zip file.

C++ evaluation will be based on: correct syntax, proper return types, proper arguments of functions, data members and class interfaces, setters/getters, unnecessary void functions, correct mathematical expression, comments throughout the code, separation of class implementations and interfaces.

Python evaluation will be based on: correct syntax, avoiding C-style loops, using Python features in general, comments throughout the notebook/scripts, labels, legends and plot styling and clarity in general.

Part 1 – C++

Implement an abstract class `DataSet` to represent and manipulate N real numbers $\{x_1, \dots, x_N\}$. The class must provide the following.

- A method to calculate the median $\bar{x}$.
- A pure virtual method to calculate the mean μ .
- A pure virtual method to calculate the standard deviation σ .
- A `print()` method to display on screen $\bar{x}$, μ , and σ .

Finally, provide a class `Arithmetic` that inherits from `DataSet` and:

- specifies the two pure virtual methods according to

$$\mu = \frac{1}{N} \sum_{i=1}^N x_i$$
$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2};$$

- overloads the multiplication operator (`*`) returning an `Arithmetic` instance obtained by multiplying two instances of `Arithmetic` element by element ($z_i = x_i * y_i$, for all i 's).

Your submitted material must include a file `app.cpp` that showcases the classes you implemented.

Part 2 – Python

Use a Python notebook or Python scripts to complete the following tasks.

1. Load the dictionary stored in the file `data.pickle`.
2. Plot the data content of the dictionary.
3. Set up a Monte Carlo with `emcee` to fit the data.
4. Display and comment the results of your fit.